XHMQV: Better Efficiency and Stronger Security for Signal's Initial Handshake based on HMQV

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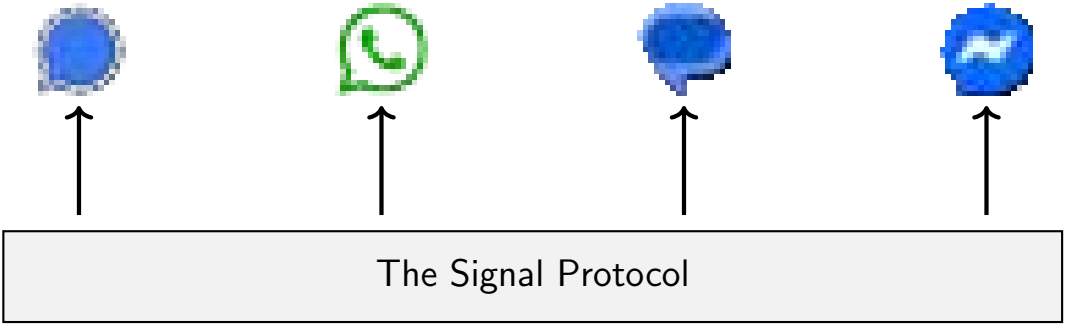
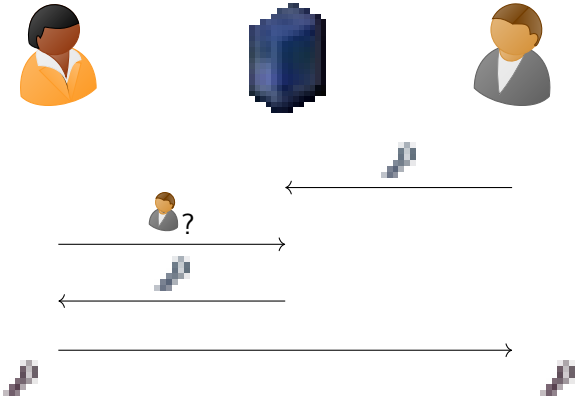
The Signal Protocol



Signal: Asynchronous Authenticated Key Exchange



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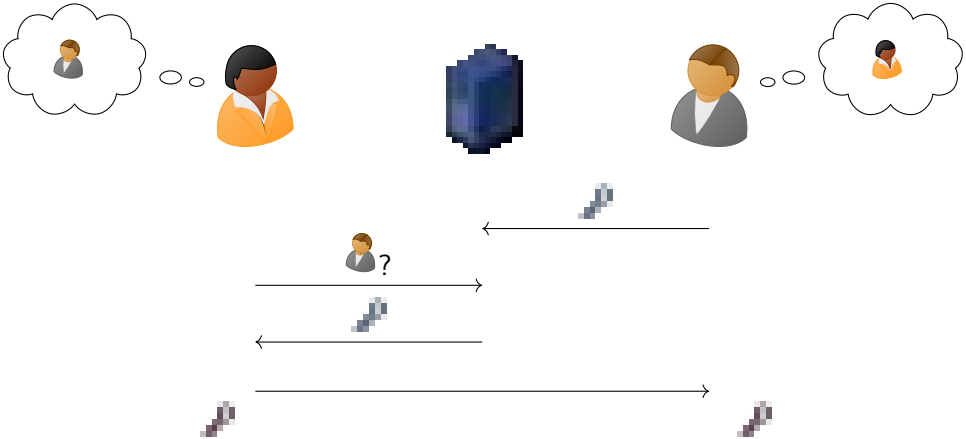

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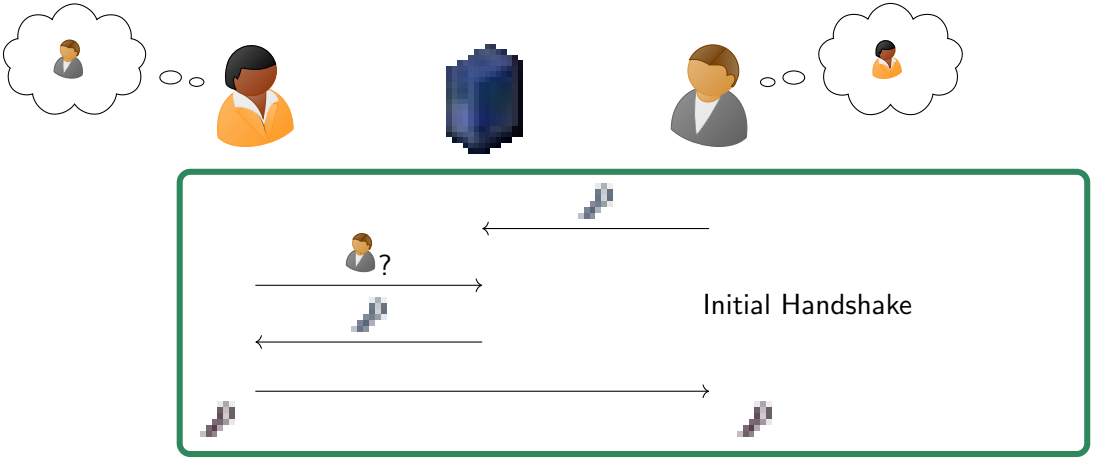

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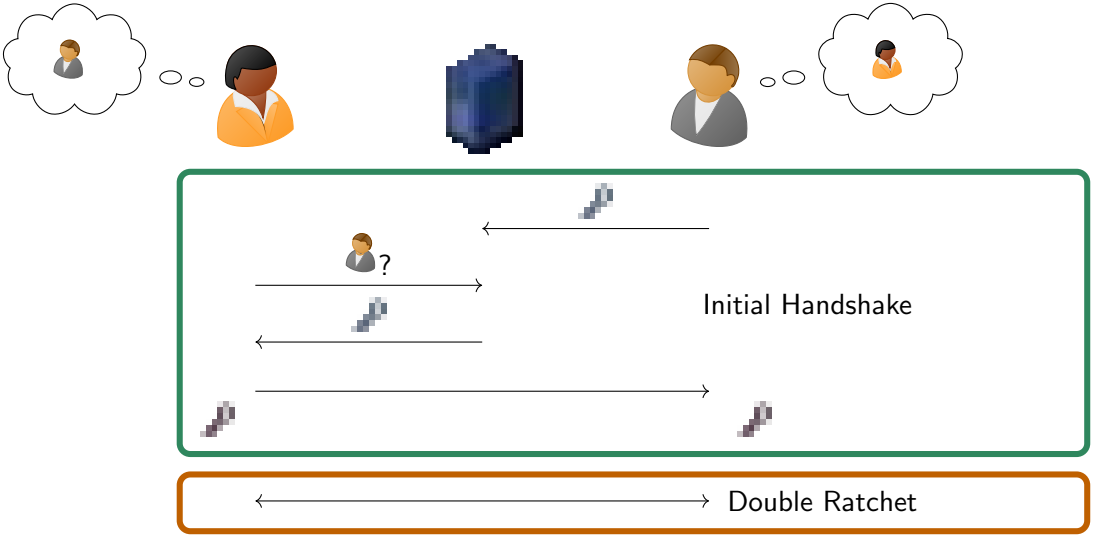

X3DH: Signal's Initial Handshake

long-term (A, a)

ephemeral (X, x)

(B, b) long-term

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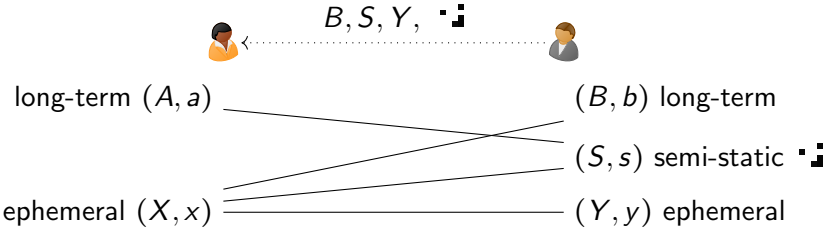

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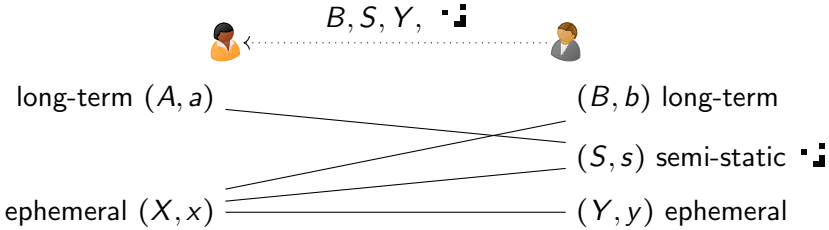

$\mathrm{KDF}(S^a \| B^x \| S^x \| Y^x)$

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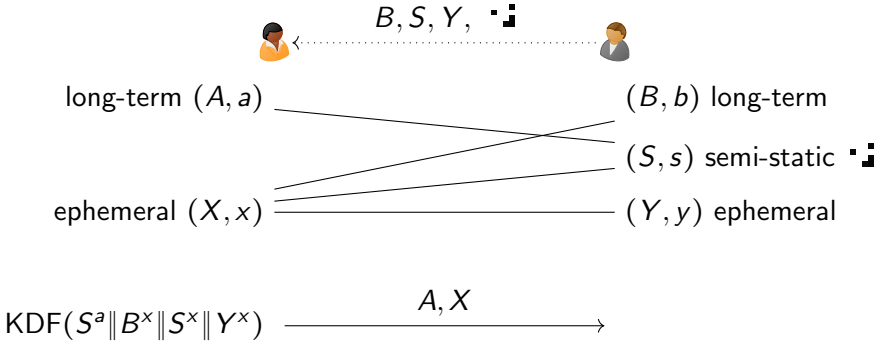

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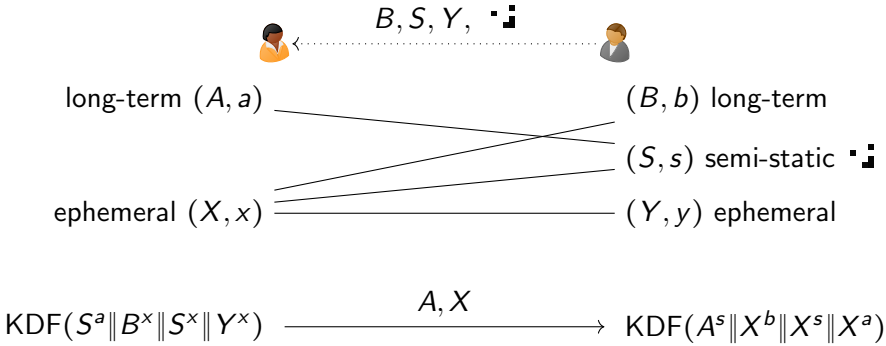
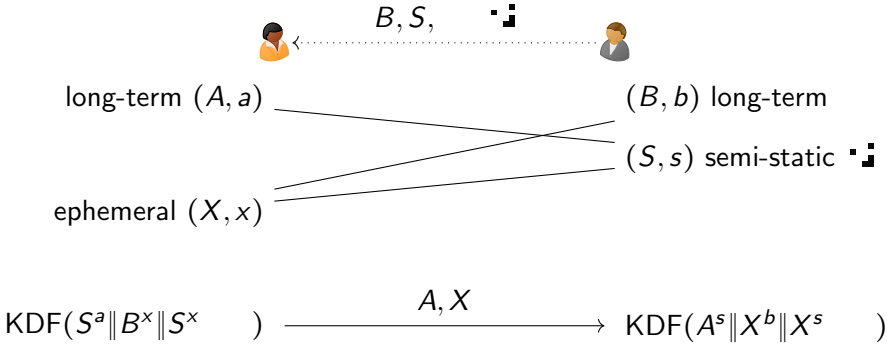


X3DH: Signal's Initial Handshake (Reduced Mode)



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Can we avoid the extra group exponentiation? $\Rightarrow$ Inspiration from HMQV [Kra05]

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- Stronger "maximum-exposure" security ✓
- No support for exhaustion of ephemeral keys ✗

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- No support for exhaustion of ephemeral keys ✗ $\Rightarrow$ Add a semi-static key to HMQV

Overview of Our XHMQV

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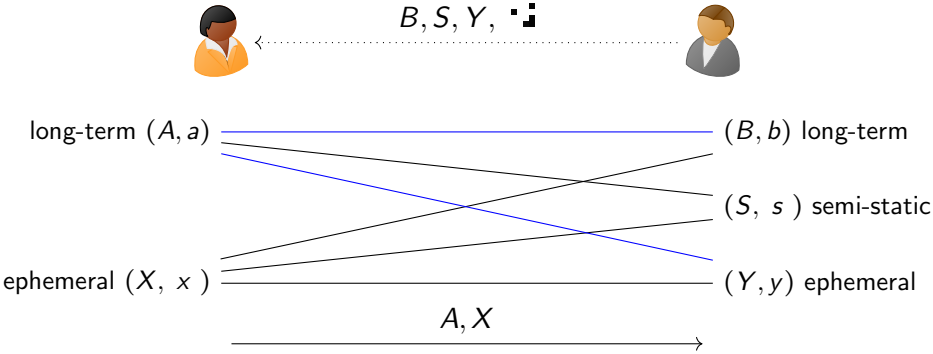

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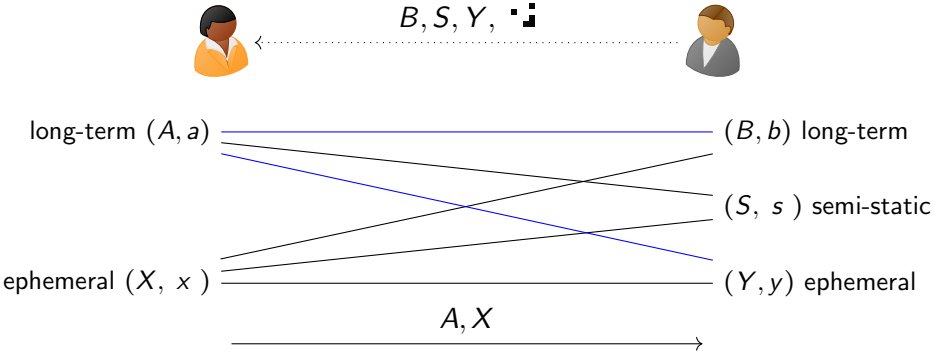

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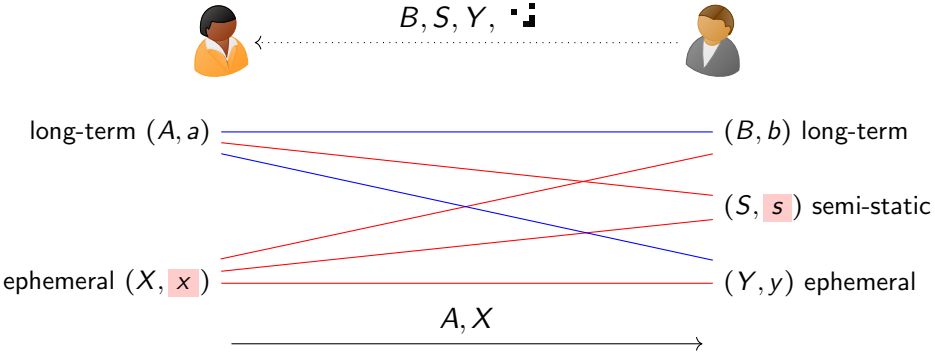

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Security of XHMQV

Security and Efficiency Comparison

Scheme	#Exp	Combinations contributing to security						Analysis	Key reuse?
		LT-SS	E-LT	E-SS	E-E	LT-LT	LT-E		
X3DH	8 (6)	✓	✓	✓	(✓)	✗	✗	[CCD+17]	✗
XHMQV	5 (4)	✓	✓	✓	(✓)	✓	(✓)	[FGPZ25]	✓
MuDH	5 (4)	✓	✓	✓	(✓)	✗	✗	[BCFS26]	✗

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Scheme	semi-honest adversaries	malicious adversaries w/ Internet access
	Signal wants this	
X3DH	✓ [FJ24]	✗ [FL25]
XHMQV	✓	✗

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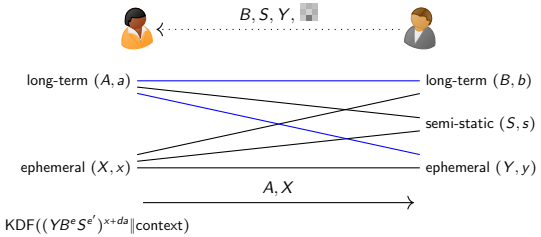
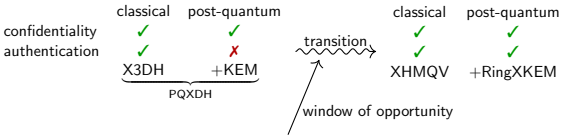
	classical	post-quantum		classical	post-quantum
confidentiality	✓	✓	transition	✓	✓
authentication	✓	✗		✓	✓
	X3DH	+KEM		XHMQV	&RingXKEM
	PQXDH				

window of opportunity

XHMQV: better efficiency & security – to be deployed in Signal





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